

Table 1: Extension 1 — The Safe-Asset Cycle: VIX Sensitivity of Dark Matter Exports (1990–2022)

	(1)	(2) Interact.	(3) FE	(4) FE+Int.	(5) Two-way
(Intercept)	-1.4390**	-1.4249**			
	(0.6641)	(0.6619)			
$\Delta \log(\text{VIX})$	4.5027*	7.3394***	4.6037	7.4364*	
	(2.5874)	(2.7980)	(3.6254)	(3.9241)	
FDI liabilities / GDP	-0.0070	-0.0071	-0.0021	-0.0022	-0.0030
	(0.0134)	(0.0134)	(0.0091)	(0.0090)	(0.0095)
FDI assets / GDP	0.0034	0.0036	-0.0002	-0.0001	0.0008
	(0.0145)	(0.0144)	(0.0065)	(0.0064)	(0.0070)
Country FE	No	No	Yes	Yes	Yes
Year FE	No	No	No	No	Yes
$\Delta \log(\text{VIX}) \times \text{Safe haven}$		-29.3406***		-29.5249**	-29.2679**
		(7.6185)		(11.3907)	(11.5328)
$\Delta \log(\text{VIX}) \times \text{OPEC}$		0.9565		1.2427	1.4939
		(12.5956)		(9.1032)	(9.2819)
Num.Obs.	2420	2420	2420	2420	2420
R2	0.003	0.008	0.055	0.061	0.095
FE: iso3c			X	X	X
FE: year					X

Notes: Dependent variable: annual dark matter exports divided by GDP. Safe-haven countries: USA, CHE, DEU, GBR, JPN, NLD, AUT, DNK, NOR (Maggiori 2017). Columns (1)–(2): heteroskedasticity-robust standard errors. Columns (3)–(5): standard errors clustered by country. In column (5), year fixed effects absorb the aggregate VIX level, so only the differential response of safe havens relative to other countries is identified. A positive and significant interaction confirms that H&S's insurance mechanism is countercyclical, not merely a cross-sectional regularity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.